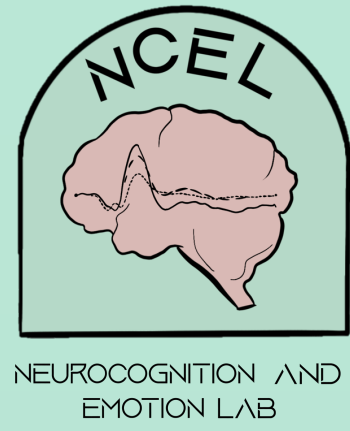


From Trauma to Comorbidity: Physical and Psychological Health in Adult Offspring of 9/11-Exposed Parents

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Introduction

- Research on the intergenerational transmission of trauma (ITT) has focused primarily on psychiatric outcomes in offspring, with substantial evidence linking parental trauma exposure to offspring PTSD, depression, anxiety, and other psychological difficulties. (Lambert, 2014; Dashorst, 2019)
- Trauma exposure in directly affected individuals is associated not only with psychiatric disorders but also with adverse physical health outcomes, including pain (Ravyts, 2026), cardiovascular (Giesinger, 2020), gastrointestinal (Li, J. 2011), and other chronic health problems.
- Despite evidence that parental trauma may have consequences across both psychological and physical domains, considerably less is known about its association with physical health in offspring, and particularly about the co-occurrence of physical–psychiatric conditions.

Objectives

- To examine whether parental occupational exposure during the 9/11 response was associated with physical–psychiatric comorbidity among adult offspring of WTC responders with lifetime PTSD.
- To determine whether associations between parental occupational exposure and offspring physical-psychiatric comorbidity differed according to offspring developmental age at the time of 9/11.

Statistical Analysis

- Multinomial logistic regression models examined the association between parental occupational exposure during the 9/11 response and offspring-reported post-9/11 experiences.
- Models were adjusted for offspring age at 9/11, offspring sex, and parent sex; offspring age at 9/11 was also examined as a moderator.
- Family-level cluster bootstrap inference accounted for multiple offspring from the same responder family while retaining all offspring in the analysis.
- Post hoc comparisons: Current age: Physical only > Neither (adjusted $p = .018$), Physical only > Psychiatric only (adjusted $p = .014$); age at 9/11: Physical only > Neither (adjusted $p = .045$), Physical only > Psychiatric only (adjusted $p = .022$); child sex: Comorbid vs. Neither (adjusted $p = .032$).

Results

- Of all the parental exposure variables, exposure to human remains was associated with greater odds of offspring comorbidity (aOR = 2.55, family-cluster bootstrap 95% CI [1.21, 6.33], $p = .023$), as shown in Figure 2.
- Associations between parental occupational exposures and offspring health varied by offspring age at 9/11. Parental debris-pile work was associated with greater odds of comorbidity among younger offspring (age 4: OR = 4.28, $p = .002$), but not older offspring (Figure 3).
- Some occupational exposures, including dust exposure, showed unexpected inverse associations with offspring health. Specifically, children of parents with greater dust exposure were less likely to have psychiatric or comorbid health conditions (Figure 4).

Figure 2. Probability of Offspring Comorbidity by Parental Exposure to Human Remains

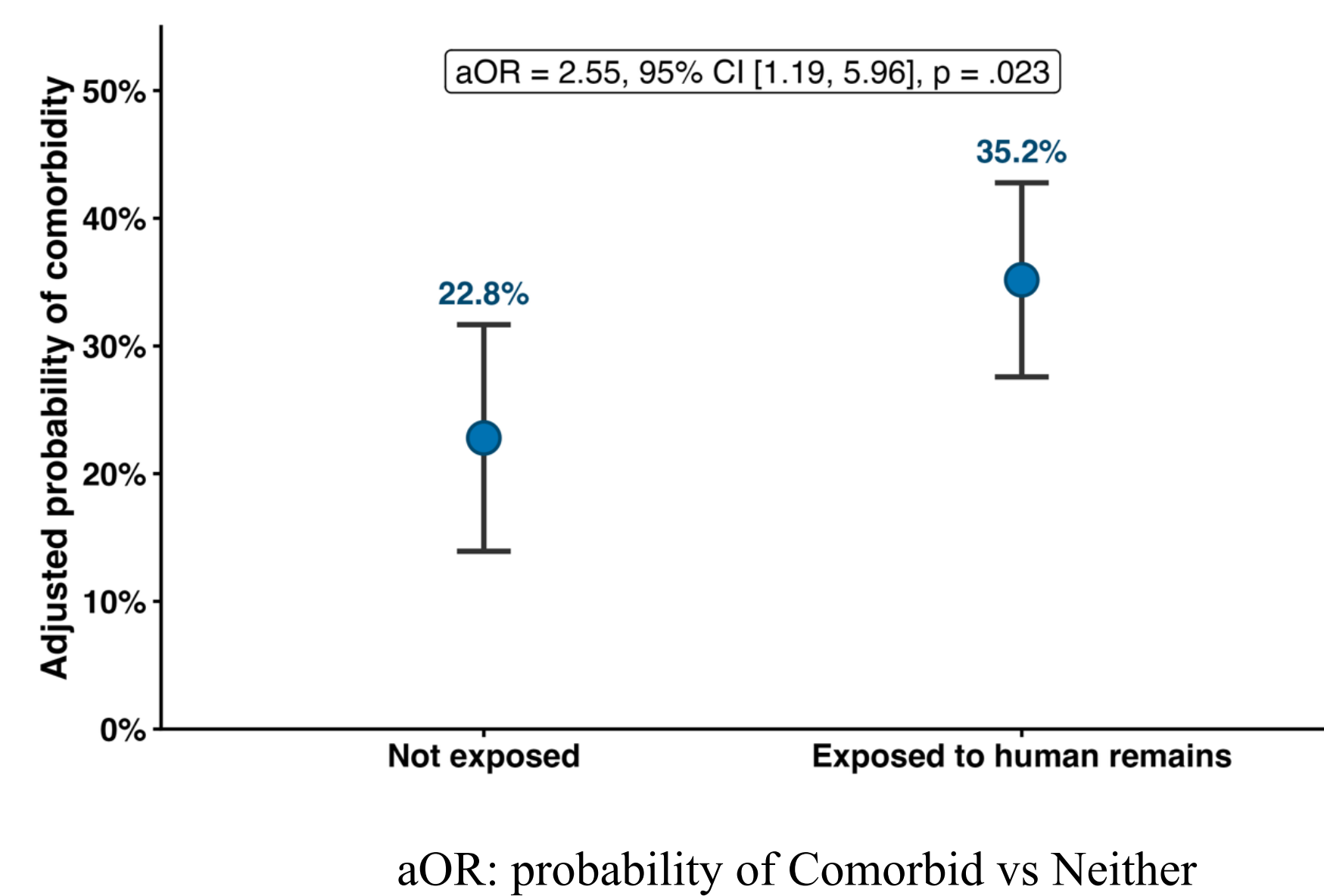


Figure 3. Probability of Offspring Comorbidity by Parental Exposure to Debris-Pile Work and Offspring Age at 9/11

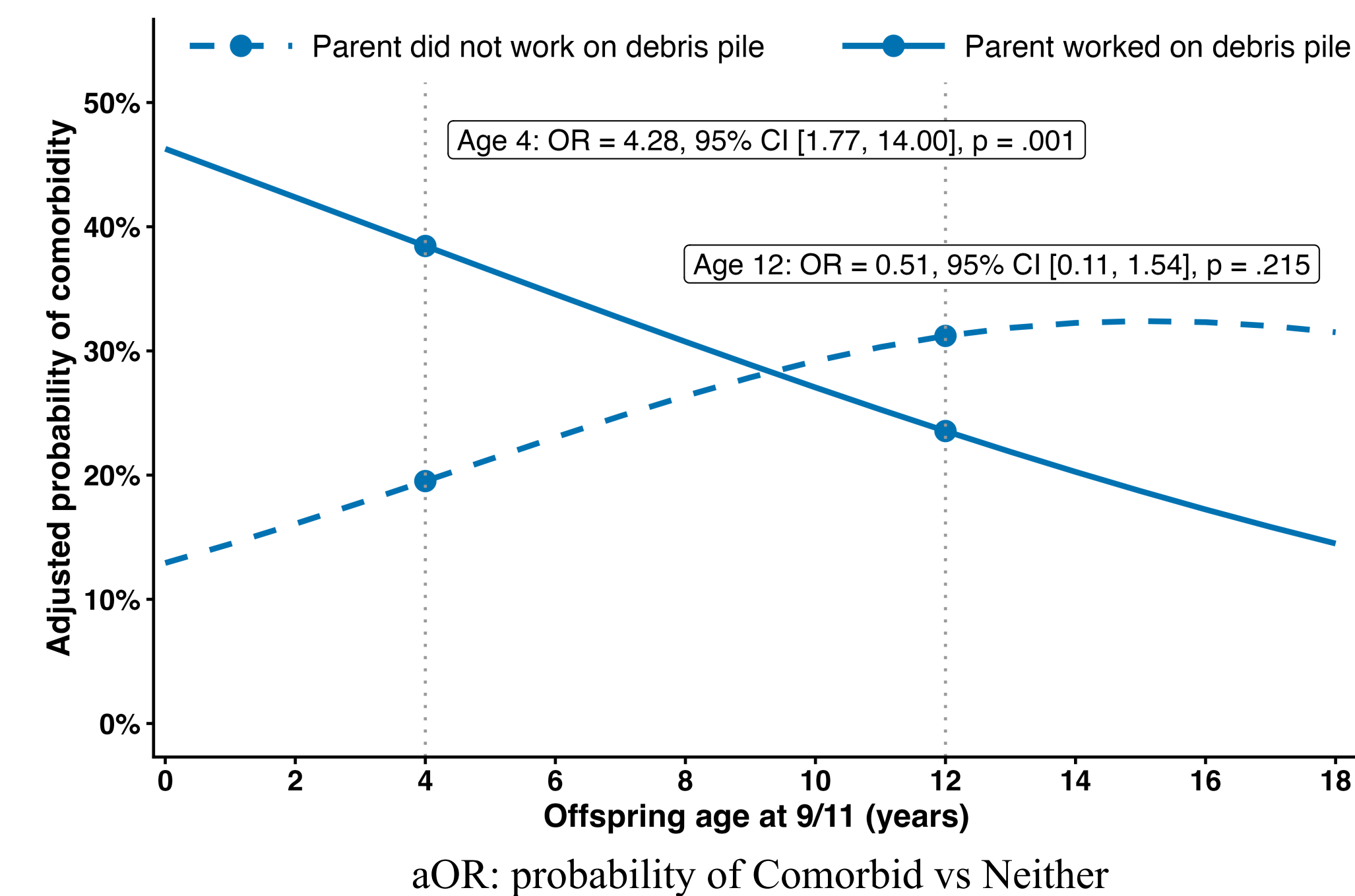
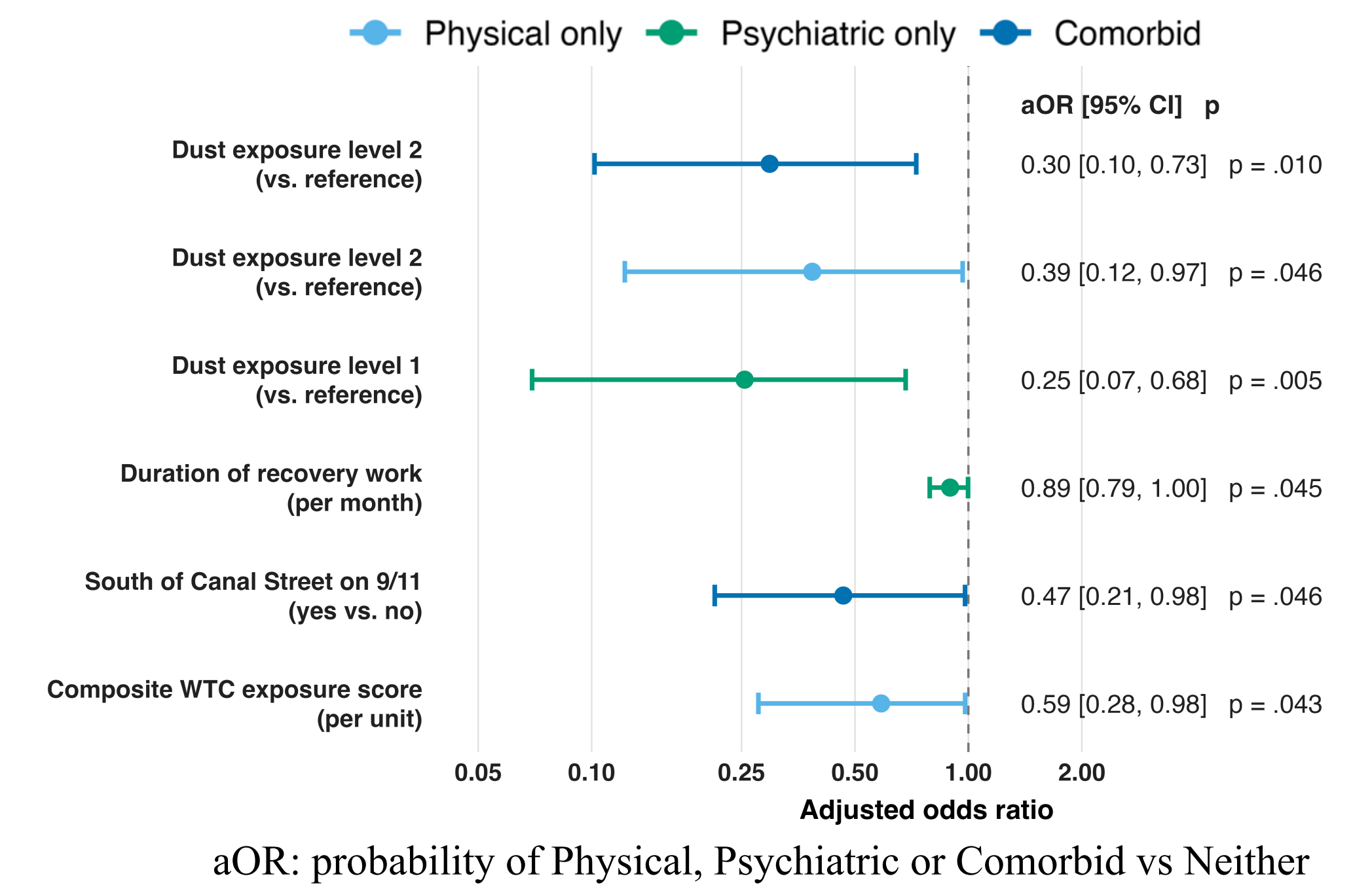


Figure 4. Association Between Parental Occupational Exposures and Offspring Health Outcomes



Study Procedures

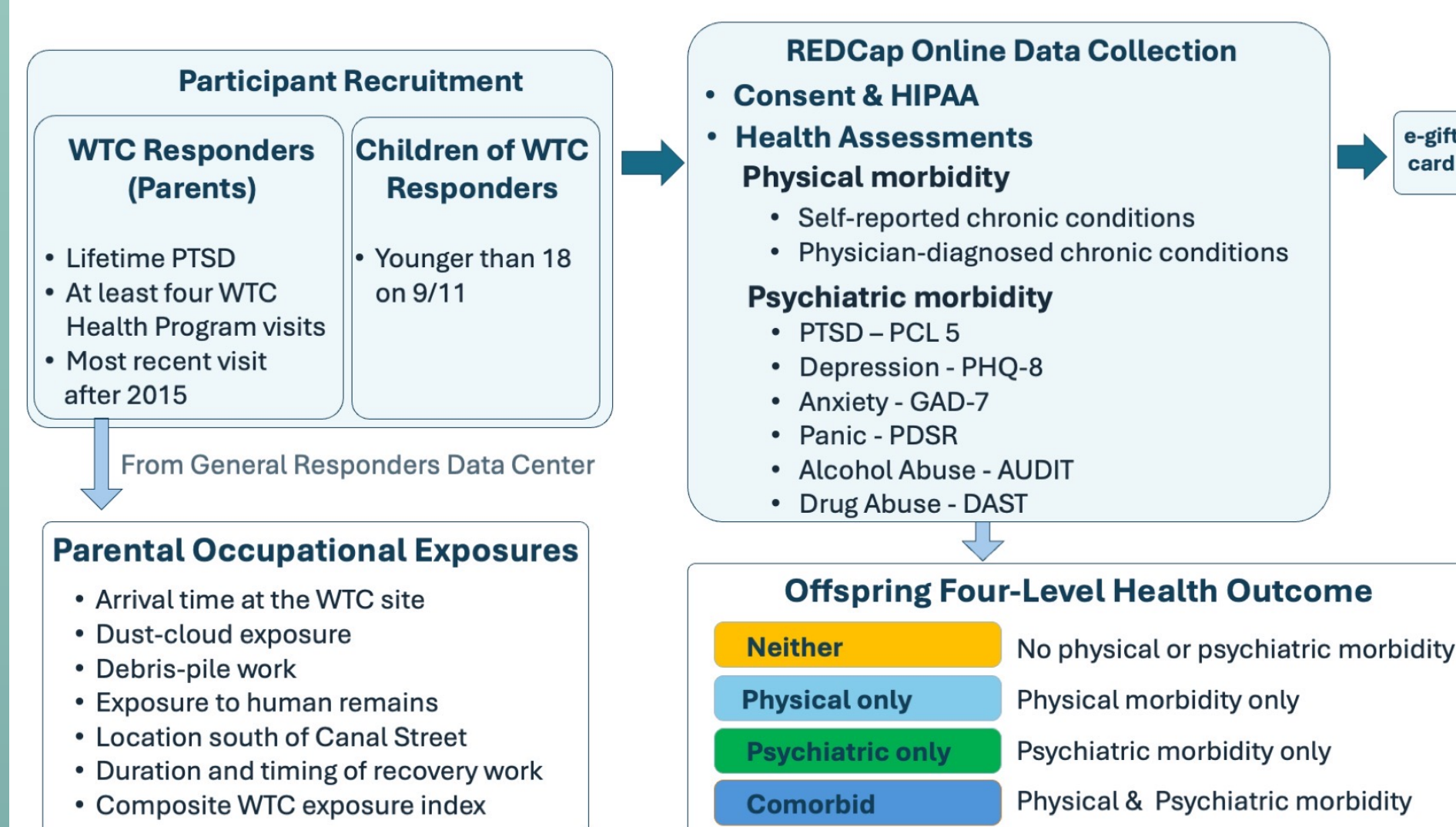
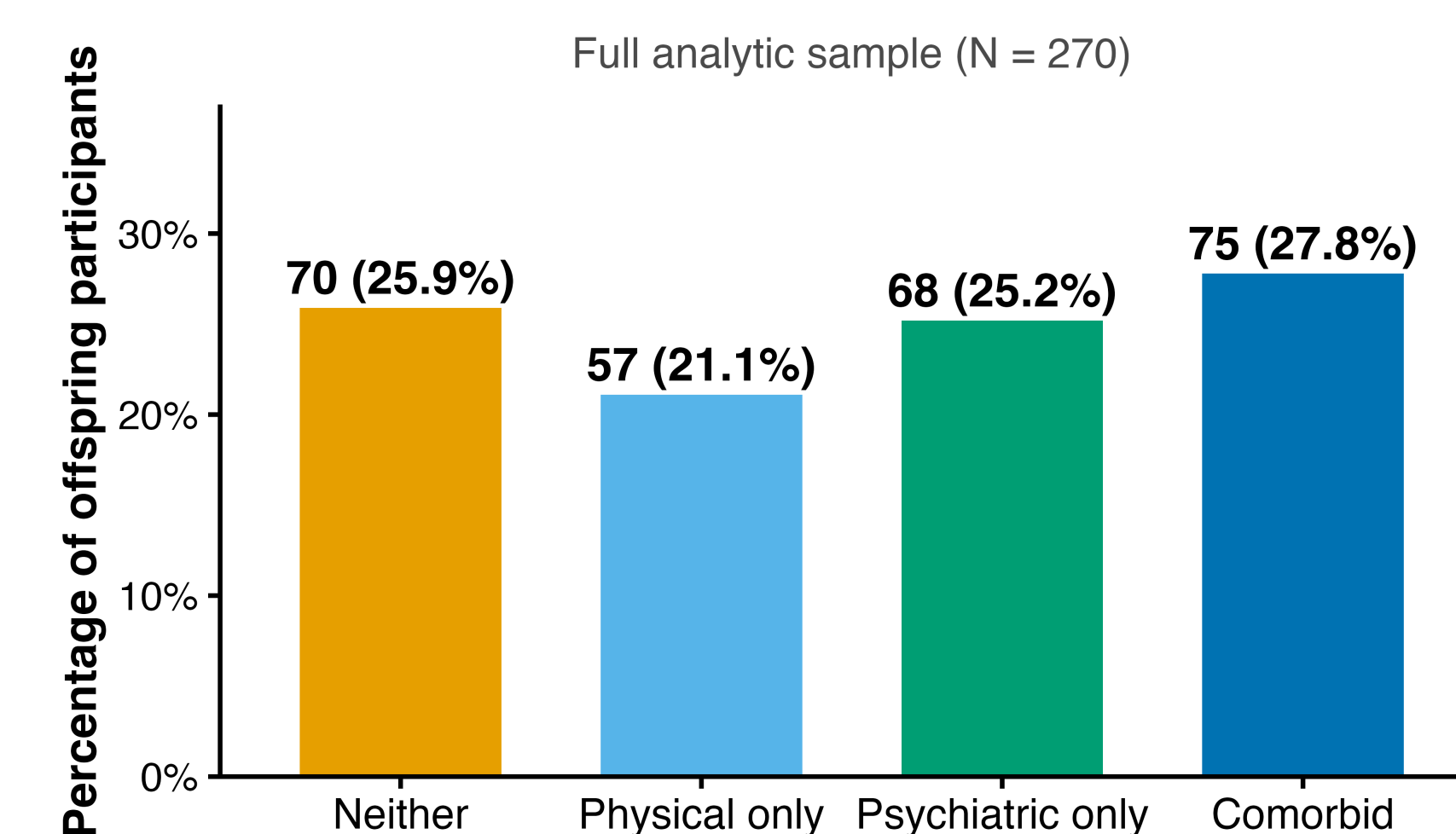


Table 1: Sample Demographics

Characteristic	Overall (N=270)	Neither (N=70)	Physical only (N=57)	Psychiatric only (N=68)	Comorbid (N=75)
Current Age, years					
Mean (SD)	28.7 (5.1)	28.1 (5.2)	30.5 (5.4)	27.9 (5.1)	28.6 (4.6)
Range	18.0-43.0	19.0-43.0	18.0-41.0	19.0-38.0	20.0-41.0
Age at 9/11, years					
Mean (SD)	8.1 (4.8)	7.4 (4.8)	10.0 (4.9)	7.3 (5.0)	7.9 (4.4)
Range	0.0-18.0	0.0-17.0	0.0-18.0	0.0-17.0	0.0-16.0
Child sex					
Male	98	31 (31.6%)	23 (23.5%)	26 (26.5%)	18 (18.4%)
Female	166	36 (21.7%)	34 (20.5%)	39 (23.5%)	57 (34.3%)
Other/Unknown	5	3(60.0%)	0(0.0%)	2(40.0%)	0(0.0%)
Spanish-speaking household					
Spanish speaking	78	22 (28.2%)	19 (24.4%)	14 (17.9%)	23 (29.5%)
Non-Spanish speaking	186	46 (24.7%)	36 (19.4%)	52 (28.0%)	52 (28.0%)
Race/ethnicity					
White	229	60 (26.2%)	48 (21.0%)	58 (25.3%)	63 (27.5%)
Minoritized Groups	40	9 (22.5%)	9 (22.5%)	10 (25.0%)	12 (30.0%)

Figure 1. Distribution of Offspring Health Outcomes



References

- Amsel, L., Cykowicz, Y. M., Rodriguez-Moreno, D. V., Cheslack-Postava, K., McReynolds, L. S., Musa, G. J., & Hoven, C. W. (2024). The long-term physical–psychiatric comorbidities related to childhood exposure to 9/11 trauma. *International Journal of Environmental Research and Public Health*, 21(8), 988–988. <https://doi.org/10.3390/ijerph21080988>
- Cyowicz, Y. M., Rodriguez-Moreno, D. V., Craft, D., & Cheslack-Postava, K. (2026). The long shadow of 9/11: Mental health outcomes in adult children of World Trade Center Responders with PTSD. *PLOS Mental Health*, 3(5), e0000574. <https://doi.org/10.1371/journal.pmen.0000574>
- Dashorst, P., Mooren, T. M., Kleber, R. J., de Jong, P. J., & Huntjens, R. J. C. (2019). Intergenerational consequences of the holocaust on offspring Mental Health: A systematic review of associated factors and mechanisms. *European Journal of Psychotraumatology*, 10(1), 1654065. <https://doi.org/10.1080/20008198.2019.1654065>
- Lambert, J. E., Holzer, J., & Hasbun, A. (2014). Association between parents' PTSD severity and children's psychological distress: A meta-analysis. *Journal of Traumatic Stress*, 27(1), 9–17. <https://doi.org/10.1002/jts.21891>
- Li, J., Brackbill, R. M., Stellman, S. D., Farfel, M. R., Miller-Archie, S. A., Friedman, S., Walker, D. J., Thorpe, L. E., & Cone, J. (2011). Gastroesophageal reflux symptoms and comorbid asthma and posttraumatic stress disorder following the 9/11 terrorist attacks on World Trade Center in New York City. *American Journal of Gastroenterology*, 106(11), 1933–1941. <https://doi.org/10.1038/ajg.2011.300>
- Ravyts, S. G., Hall, R., Vandine, D., Woods, O., Harris, R., Dudeney, J., & Aaron, R. V. (2026). PTSD and chronic pain: A systematic review and meta-analysis of mediation studies. *The Journal of Pain*, 38, 105589. <https://doi.org/10.1016/j.jpain.2025.105589>
- Zhou, M., Larsson, H., D'Onofrio, B. M., Landén, M., Lichtenstein, P., & Pettersson, E. (2023). Intergenerational transmission of psychiatric conditions and psychiatric, behavioral, and psychosocial outcomes in offspring. *JAMA Network Open*, 6(12), e2348439. <https://doi.org/10.1001/jamanetworkopen.2023.48439>

Conclusions

- Parental occupational exposure during the 9/11 response was associated with physical–psychiatric health outcomes in adult offspring, with associations varying by the type of parental exposure and the offspring's age at 9/11, highlighting potential long-term and intergenerational consequences of exposure to collective trauma.
- The unexpected divergent association with parental exposure may reflect differences in the nature of the exposures. Psychologically salient exposure, such as contact with human remains, may increase parental stress and likelihood of ITT, contributing to physical–psychiatric comorbidity in offspring. In contrast, primarily physical exposures, such as dust, may have elicited different parental responses, including greater efforts to protect children from potential harm, potentially reducing the ITT and resulting in an inverse pattern of association.
- These findings extend ITT research beyond psychiatric outcomes and highlight the importance of considering the nature of parental exposure, physical–psychiatric comorbidity, and developmental timing in understanding ITT effects.

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